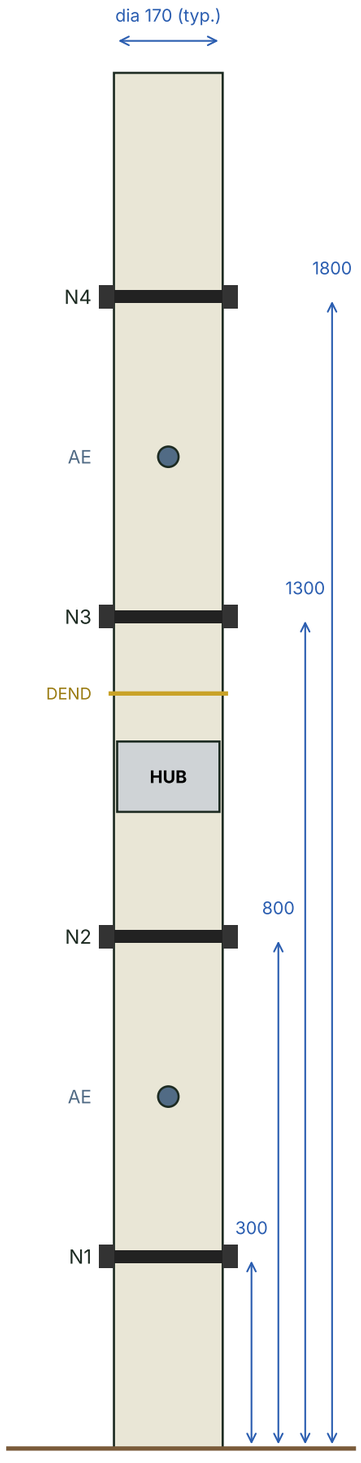
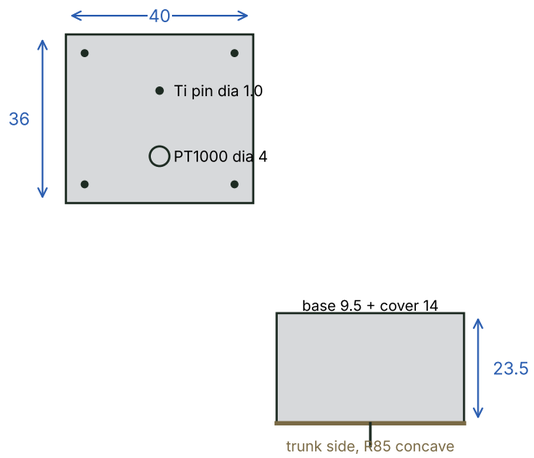
PhytoHydra v1 - mechanical drawing sheet

A Installation elevation

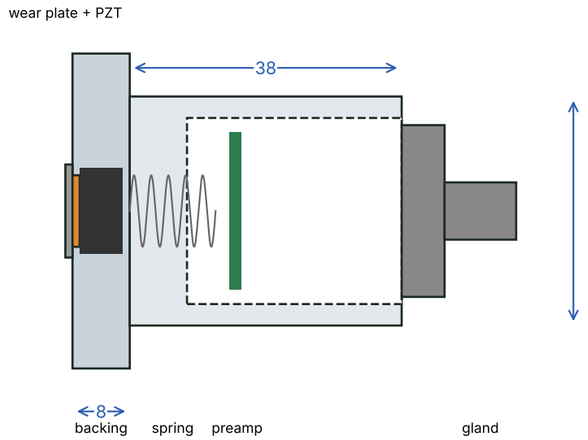


Seedling rig: N1-N4 at 50/150/250/350 mm

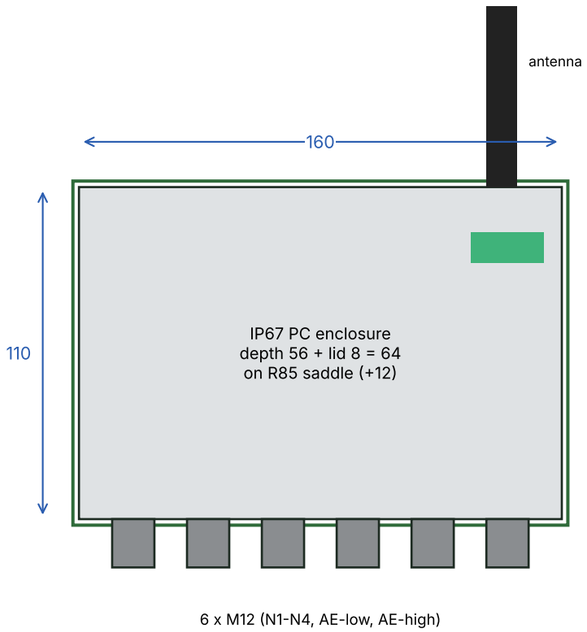
B Electrode pod (front / top section)



C Acoustic Sentinel (axial section)



D Edge Hub (front)



6 x M12 (N1-N4, AE-low, AE-high)

Item	Material / part	Key spec
Electrode pin	Ti gr.2 or SS316	dia 1.0, insertion 5-8 mm, replace yearly
Pod base + cover	ASA / PA12, 3D printed	UV stable, 4 x M2, silicone gasket
Strap	TPU 95A, over-centre buckle	20 × 2.2 mm, 2-5 N/cm2 contact
AE sensor	PZT-5A disc 10 × 1 mm	~200 kHz, SS wear plate, gel couplant
AE preload spring	SS302, OD 13, wire 0.7	10-15 N on wear plate
Dendrometer	Invar 0.3 mm band + AS5311	0.5 um resolution, spring tension 5 N
Hub enclosure	IP67 polycarbonate	160 × 110 × 64, M12 A-coded x 6
Hub saddle	ASA, 2 x SS band clamps 20 mm	R85 for 170 mm trunk (param.)
Solar	6 W mono panel on 20 mm arm	25 deg tilt, south facing
Cables	4-core shielded, PUR jacket	driven shield, max 2.5 m per band

NOTES

1. All trunk-contact radii are parametric (trunk_d in phytohydra_params.scad).
2. Excitation <10 uA RMS, <1 V, AC only, DC-blocked: no electrolysis.
3. Pods sit 180 deg apart so each band gives 2 sides (A/B).
4. AE sensors share one vertical line for arrival-time zoning.
5. Keep all cable exits pointing down (drip loops) in monsoon.
6. Mount hub on the north side of the trunk, panel on an arm clear of the frond skirt.
7. Rev A is design intent: fit, IP rating and contact force to be verified in Phase 0 bench tests.

PhytoSense AI | PhytoHydra v1 prototype

Drawing PH-ME-001 rev A | Key mechanical dimensions

Units: mm | Scale: NTS | Date: 2026-10-07 | Status: design intent, not yet built